Priyansu Biswal

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Education

Texas A&M University College Station, TX
B.S. in Computer Science (Engineering Honors) *Expected May 2028*
- **Relevant Coursework:** Program Design & Concepts, Data Structures & Algorithms, Discrete Structures & Computing

Technical Skills

Languages: Python, TypeScript, SQL, JavaScript, Java, C++
Frameworks: FastAPI, Next.js, React.js, SQLAlchemy, Pydantic, Node.js, REST APIs
AI/ML: LLM APIs (Claude, OpenRouter), RAG, PyTorch, XGBoost, YOLO11, pandas, NumPy
Databases: PostgreSQL, pgvector, Redis
Cloud/Tools: Docker, Kubernetes, Azure (Container Apps, AKS, ACR), AWS (EC2), GitHub Actions, Pytest, Playwright, Git

Experience

Team Lead – Future Market Prediction Sept. 2026 – Present
Aggie Data Science Club (American Airlines Partnership) *College Station, TX*
- Leading a team of **4 students** in a club partnership with American Airlines to forecast airline market demand up to 2 years out for its Network Planning team
- Training an XGBoost model on **300K+ flights** that currently explains **98% of ticket price variation** ($R^2 = 0.978$) on unseen flights, finding seat class and airline drive prices the most

Software Engineer Intern June 2026 – Aug. 2026
Finch *College Station, TX*
- Built backend and AI features for Finch, a job-search app with **250+ registered users** that matches users to jobs and tailors resumes with LLMs
- Built a RAG pipeline that retrieves jobs from **93K** pgvector embeddings for LLM fit scoring, filtering by seniority so intern searches stop returning senior roles; cut vector search from **8.3 s to 1.3 ms** with an HNSW index
- Sped up the first pass of the user profile builder (which turns a user's LinkedIn, GitHub, and resume into a structured profile) **2.19**× by moving it to a faster LLM, with no drop in output quality in side-by-side tests
- Cut failed profile builds from **38% to 5%** with the team by catching cut-off LLM responses, forcing structured JSON output, and splitting each project into its own parallel LLM call
- Automated releases with GitHub Actions CI/CD that tests every PR (Pytest, Playwright) and deploys to staging and production, removing manual release steps for **all 5 developers**

Quant Engineer Jan. 2026 – Present
Maroon Fund at Texas A&M *College Station, TX*
- Rebuilt the backtesting system for Maroon Fund, a student-led fund managing **$70K**, with the team after catching a **104%** backtest that had been tested on its own training data
- Backtested a **65%** return vs. **26%** for simply holding NVDA over a year of market data the model never saw, after trading costs (Sharpe ratio **1.43**)
- Added risk metrics showing the strategy caught **137%** of the S&P 500's gains on up days but only **89%** of its losses on down days

Projects

Search Engine | *Python, FastAPI, PostgreSQL, Redis, Docker, Kubernetes* May 2026 – Present
- Built a web search engine from scratch that crawls, indexes, and ranks pages with field-aware BM25, returning the right page as the top result for **90%** of test searches
- Replaced a global delay with a shared Redis frontier and per-domain limits for **5x** faster crawling (27 → 135 pages/min)
- Made saving each new page **100x faster** with a write-ahead log, so newly crawled pages become searchable **twice as fast**
- Deployed the crawler, indexing worker, and search API as independently scaling services on Azure Kubernetes Service with managed PostgreSQL and Redis, isolating failures so one crashed service can't take down the rest

FleetOS | *FastAPI, Next.js, PostgreSQL, Docker, AWS EC2, Azure AKS* Feb. 2026 – Present
- Built a trucking fleet-management backend: **40+ REST endpoints** in FastAPI on a multi-tenant PostgreSQL schema
- Deployed the API on an AWS EC2 instance and added fault-tolerant Geotab GPS and DAT load-board syncing
- Created a scheduled ETL job on Azure Kubernetes Service that validates, dedupes, and upserts **2M+ federal carrier records** into PostgreSQL, replacing a manual daily script
- Implemented a dispatch scorer that ranks trucks for each load by empty miles, remaining legal driving hours, and delay risk

Next Up | *Python, PyTorch, YOLO11, Ultralytics* May 2026 – Present
- Building a high school basketball film analytics tool, where no public models exist and courts overlap other sports' lines
- Fine-tuned a YOLO11 pose model to detect 22 court landmarks on frames labeled with a custom annotation tool, raising detection accuracy **3x** (mAP 16.6% → 51.2%)